

AI Application Engineer Job Description

Company Background

FlowAI is a small software company building internal AI tools for sales, operations, and customer support teams. The company wants to turn large language models into practical applications that solve real business problems.

Role Summary

We are looking for a junior AI Application Engineer who can build useful LLM-powered applications. This role focuses on retrieval augmented generation, agent workflows, data analysis assistants, and simple web applications.

The engineer does not need to train foundation models from scratch. The main responsibility is to connect models, data, tools, and user interfaces into reliable products.

Responsibilities

- Build RAG applications for PDF, Markdown, and internal knowledge base documents.
- Use embeddings and vector databases to retrieve relevant context.
- Design prompts that make model outputs clear, grounded, and useful.
- Build simple agent workflows that can call tools such as search, file readers, and data analysis functions.
- Create Streamlit or web-based demos for non-technical users.
- Evaluate answers by checking source documents, relevance, and hallucination risk.
- Write clean documentation so teammates can run and understand the project.

Required Skills

- Basic Python programming.
- Familiarity with LangChain or similar LLM application frameworks.
- Understanding of RAG concepts: chunking, embeddings, vector stores, retrievers, prompts, and generation.
- Basic knowledge of APIs, environment variables, and Git.
- Ability to debug errors step by step.
- Clear communication and willingness to learn.

Nice To Have

- Experience with Streamlit, FastAPI, or simple frontend development.
- Experience with Chroma, FAISS, Pinecone, or other vector databases.
- Knowledge of NLP, neural networks, and deep learning basics.
- Experience building AI agents with tool calling.
- Ability to turn business requirements into small working prototypes.

Example Interview Questions

1. What problem does RAG solve?
2. Why do we split documents into chunks?
3. What is the difference between embeddings and prompts?
4. How do you reduce hallucination in a document QA system?
5. How would you evaluate whether a retrieved document chunk is relevant?
6. How would you explain your AI application project to a non-technical user?

Project Assignment

Build a small AI Job Description Knowledge Base. The user can upload job descriptions and learning notes, then ask questions such as:

- What skills does this role require?
- Which skills are most important for a beginner?
- What should I learn first?
- What project can I build for my resume?
- What interview questions should I prepare?

The answer should cite the relevant source text when possible. If the document does not contain enough information, the system should say that it cannot find enough evidence.